

Southeastern Section - 68th Annual Meeting - 2019

Paper No. 16-4

Presentation Time: 1:00 PM-5:00 PM

TIME SERIES ANALYSIS OF VOLCANIC TEMPERATURE MONITORING: AN EXAMPLE FROM THE NISYROS ACTIVE VOLCANO AT THE HELLENIC VOLCANIC ARC

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An efficient mitigation of a volcanic natural hazard is the continuous monitoring and separation from interference. In this research, we examine the results of a decomposition analysis from a crater's surface temperature data and associated climatic variables. Nisyros active volcano located at the eastern segment of the recent Hellenic Volcanic Arc, and it has shown temperature surface oscillation that may reflect repeated heat emission from a currently active hydrothermal system that mostly is affected by the seasonal meteorological parameters and the atmospheric air. Time series from the meteorological station of Kos island have been utilized, in order to investigate the atmospheric influence to the time series of the surface temperature fluctuations of Nisyros caldera. The volcanic temperature raw data have been obtained from a temperature TinyTag Plus probe TGP-4520 that has been established at the southern part of the caldera nearby an active fumarolic emanation area. Temperature surface readings from August of 2016 to January of 2017 were analyzed and decomposed into three different components which are long-, seasonal- and short-term, respectively. For this purpose, both the Kolmogorov-Zurbenko and the Autoregressive Distributed Lag (ADL) model have been incorporated. In comparison with the multiple linear regression (MLR) model, the application of the time series regression (TSR) model and the ADL model, provided the opportunity to integrate lag operation from the dependent and the independent variables. Thus, both the coefficient determination (R^2) of the three components (long-, seasonal- and short-term) increased and the physical interpretation of the lagged values strengthened the accuracy of the predicted model. The total explanation of the predicted model was 75% and the unexplained part was 25%, whereas the unexplained part of the raw data was 66%. Therefore, the utilized methodology has increased the overall explanation of the model's forecast by more than three times. The decomposition of the temperature readings and isolating the atmospheric factor may allow an in-depth understanding of the heat circulation and emission of the hydrothermal system underneath the volcanic caldera surface.

Session No. 16--Booth# 4

[T12. Natural Hazards Prediction \(Posters\)](#)

Thursday, 28 March 2019: 1:00 PM-5:00 PM

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